Critical Evaluation and Paradigm Assessment: Transport-Aligned Heterogeneous Observer-Routed Neighborhood Attention (THORN)

1. Paradigm Evaluation and Conceptual Novelty

The manuscript proposing the Transport-Aligned Heterogeneous Observer-Routed Neighborhood Attention (THORN) architecture presents a highly ambitious synthesis of topological analysis and representation learning. The prevailing paradigm in Graph Neural Networks (GNNs) and Graph Transformers has predominantly relied on content-based routing, wherein attention weights are derived exclusively from the learned query-key projections of semantic node features. In such architectures, the underlying graph structure is relegated to a passive role, typically functioning merely as a binary mask that restricts the attention neighborhood to physically connected nodes. The THORN architecture fundamentally challenges this status quo by elevating topological structure from a static constraint to a dynamic, learned routing prior. By utilizing manifold-geometric descriptors as active gating signals within a Mixture-of-Experts (MoE) router, the framework proposes a definitive shift in how structural biases are integrated into attention mechanisms.

A critical evaluation of this conceptual leap requires distinguishing between an incremental architectural modification and a genuine paradigm shift. The integration of geometric features into GNNs is not inherently novel; established models such as Curvphormer utilize discrete Ricci curvature as a structural bias directly added to attention weights to mitigate over-squashing. Similarly, Ollivier-Ricci Curvature-Aware Graph Convolutional Networks (ORCA-GCN) deploy curvature to reweight message passing, soft-pruning redundant intra-cluster edges while amplifying low-curvature bridge edges. However, these established methods still operate entirely within a single-view topological space, treating the geometry as a static weight modifier. The genuine novelty of THORN lies in its decoupling of geometric analysis from message-passing execution, utilizing the geometry to select entirely different topological views. This decoupling represents a paradigm shift from "learning on a fixed graph" to "learning which graph view is optimal for a specific topological neighborhood." The authors successfully identify a critical gap in multi-view graph learning: forcing a single set of attention heads to process wildly divergent structural perspectives (e.g., k-Nearest Neighbors versus multi-scale diffusion) creates unavoidable representation conflicts. By routing specific attention heads to distinct graph views based on the local geometric environment, THORN implements a topology-guided MoE framework. However, to solidify this paradigm, the authors must address probing questions regarding the redundancy of their chosen geometric features. A robust theoretical paradigm requires proving that the selected geometric observers offer mutually orthogonal information rather than highly correlated topological signals.

To refine this conceptual foundation, the manuscript must mathematically formalize the covariance between the geometric descriptors before they enter the gating network. If the features are highly collinear, the MoE router's capacity is wasted on resolving redundant signals,

undermining the elegance of the proposed paradigm. The authors are advised to query academic databases for literature on "manifold redundancy" and "geometric feature orthogonalization" to ground their observer selection in rigorous information theory. Furthermore, the reliance on an offline-computed Fused Gromov-Wasserstein (FGW) alignment introduces a severe scalability bottleneck that threatens to render the paradigm empirically impractical for massive, dynamic datasets. These vulnerabilities must be systematically addressed to meet the rigorous publication standards of top-tier AI/ML journals.

2. Theoretical Foundations of Geometric Observer Routing

The utilization of geometric observer features to drive the MoE gating network is the most mathematically rich component of the THORN architecture. A rigorous evaluation must dissect the theoretical validity of each chosen metric and its appropriateness as an attention-routing signal. The manuscript currently aggregates disparate geometric metrics without sufficiently formalizing their theoretical interplay. It is imperative to dissect these metrics individually to evaluate their capacity to drive expert routing.

2.1 Ollivier-Ricci Curvature: Topologies of Over-squashing and Over-smoothing

The Ollivier-Ricci Curvature (ORC) is a discrete adaptation of the classical Ricci curvature from differential geometry, explicitly defined for metric measure spaces. In the context of graph theory, ORC quantifies the degree to which the neighborhoods of adjacent vertices overlap, embedding profound geometric insight into discrete structures. The metric utilizes the Wasserstein-1 (Earth Mover's) distance to evaluate the transportation cost between the probability distributions of random walks originating from two connected nodes. By computing this optimal transport distance, ORC provides a robust indicator of local topological density and dispersion.

In contemporary GNN literature, ORC has been intimately linked to the dual, pervasive problems of over-smoothing and over-squashing. Positive ORC values typically indicate densely connected regions or intra-community edges, which are highly susceptible to over-smoothing as repeated message passing rapidly homogenizes node representations. Conversely, highly negative ORC values indicate structural bottlenecks or bridge edges connecting distinct communities, which force vast amounts of information through narrow topological pathways and lead to severe over-squashing. THORN leverages ORC not merely to reweight these edges, but to route attention entirely away from problematic topologies.

If a node is situated in a high-curvature region susceptible to over-smoothing, the THORN MoE router can dynamically deprioritize the standard topological view. It can instead route the attention head to a feature-based kNN view or a multi-scale diffusion view that bypasses the local density trap. The theoretical grounding here is robust, but the manuscript must address the intense computational overhead of exact ORC calculations. Computing the exact Wasserstein distance for every edge requires solving a linear programming problem, which scales prohibitively on massive networks. The report should critically evaluate whether THORN should adopt an approximation, such as Forman-Ricci curvature, which provides computational advantages while empirically correlating with ORC.

2.2 Local Intrinsic Dimension: Manifold Complexity as a Routing Prior

The inclusion of Local Intrinsic Dimension (LID) provides the routing network with an estimation of the effective dimensionality of the data manifold near a specific target point. Derived through maximum likelihood estimation based on nearest-neighbor distances, LID characterizes the minimal number of variables required to represent the local topological space without significant information loss. A high LID indicates that the current point lies in a highly complex, potentially noisy local manifold that cannot be easily compressed. In the context of THORN, this provides a profound structural signal regarding feature-space reliability.

By feeding LID into the MoE router, THORN allows attention heads to adaptively seek out views that provide lower-dimensional, cleaner signals when the feature-space LID is excessively high. For example, if the kNN feature space exhibits a high LID, the router can shift attention mass to the temporal view, isolating a lower-complexity signal. This represents a highly sophisticated mechanism for dynamic noise filtering. The authors should structurally improve their hypothesis by explicitly defining the mathematical relationship between LID gradients and routing logits.

2.3 Local Outlier Factor: Density Discrepancies and Anomaly Gating

The Local Outlier Factor (LOF) quantifies the relative neighborhood density of a point compared to its immediate neighbors, traditionally serving to identify anomalous connectivity patterns. While LOF is predominantly utilized for isolated anomaly detection tasks, its application in THORN as a continuous routing prior is highly innovative. If an edge exhibits a severe LOF score, it suggests a profound density mismatch between the connected nodes, indicating a potential topological hallucination or noisy edge injection.

In the presence of high LOF scores, the THORN router can rely on cross-view consensus rather than trusting a single anomalous topology. This transforms LOF from a binary classification tool into a continuous confidence metric for structural reliability. To strengthen this theoretical paradigm, the authors must clarify how the highly non-linear LOF metric is normalized before passing through the router's layer normalization, as extreme outlier scores could destabilize the MoE gating network.

2.4 Synergy and Orthogonality of Observer Features

The primary challenge with integrating LID, ORC, and LOF simultaneously is the severe potential for manifold redundancy. The manuscript must mathematically justify that the gating network requires all three descriptors to achieve optimal routing. A Principal Component Analysis (PCA) or feature attribution study on the learned routing weights should be mandated to demonstrate that the router actively utilizes the orthogonal components of these geometric observers.

Geometric Observer	Primary Topological Measurement	GNN Pathology Addressed	Proposed THORN Routing Behavior
Ollivier-Ricci Curvature	Neighborhood overlap; optimal transport distance	Over-smoothing (dense) & Over-squashing (bottlenecks)	Routes away from bottlenecks; utilizes diffusion for dense clusters.
Local Intrinsic Dimension	Effective manifold dimensionality; local	Feature noise; high-dimensional	Shifts attention to lower-dimensional

Geometric Observer	Primary Topological Measurement	GNN Pathology Addressed	Proposed THORN Routing Behavior
	complexity	entanglement	views (e.g., temporal).
Local Outlier Factor	Relative neighborhood density; anomaly detection	Topological noise; false edge injections	Forces cross-view consensus; deprioritizes anomalous edges.

3. Differentiable Log-Barrier Masking in Graph Attention

THORN's architectural execution hinges critically on the continuous relaxation of sparse attention via log-barrier masking. This mechanism addresses a long-standing mathematical limitation in the training of sparse neural architectures over irregular domains. The authors must rigorously contextualize this advancement against historical masking techniques to prove its superiority.

3.1 Moving Beyond Hard Masking and Straight-Through Estimators

A historical limitation of standard Graph Attention Networks (GAT) is the strict reliance on hard neighborhood masks to enforce sparsity. In standard GAT, the attention softmax is mathematically confined to nodes $j \in \mathcal{N}(i)$, utilizing index-based gathering operations. This hard indexing is fundamentally non-differentiable with respect to the graph topology itself, preventing the network from utilizing backpropagation to learn which structural connections should exist. Consequently, the graph structure remains a rigid, unlearnable constraint throughout training.

Prior attempts to circumvent this non-differentiability have frequently relied on the Straight-Through Estimator (STE) to backpropagate through binary or hard-selected gates. While STEs allow for empirical training of sparse connections, they compute biased gradients by ignoring the true derivative of the thresholding function, which can lead to suboptimal optimization trajectories. THORN's log-barrier masking provides a mathematically sound alternative that maintains end-to-end gradient flow without the need for biased approximations. By rendering the mask fully differentiable, THORN allows the downstream task loss to directly optimize the upstream view selection and geometric routing.

3.2 Mathematical Formulation and Continuous Relaxation

THORN introduces a log-barrier term to gracefully enforce sparsity while preserving differentiability. The attention logits incorporate the routed edge weight $\tilde{A}_{ij}^{(i,h)}$ through the explicit additive term: $\log(\tilde{A}_{ij}^{(i,h)} + \epsilon)$. This formulation acts as an elegant continuous relaxation of discrete sparse attention, mirroring techniques used in constrained convex optimization.

In mathematical optimization, a log-barrier function imposes an increasingly severe, asymptotic penalty as a point approaches the boundary of the feasible region. By embedding this barrier directly within the softmax argument, THORN ensures that if the MoE router assigns a routing weight approaching zero to a particular view, the corresponding adjacency $\tilde{A}_{ij} \to 0$. Consequently, the logarithm approaches $-\infty$, and the softmax function assigns

mathematically vanishing probability mass to that edge. Proposition 1 of the manuscript elegantly proves that this mechanism is strictly differentiable with respect to the routing logits, solidifying the theoretical ground support for the model.

3.3 Optimization Stability and Numerical Fragility

Despite the theoretical elegance of the log-barrier formulation, it introduces acute numerical vulnerabilities that the authors must address. Log-barrier functions are notoriously sensitive to input scaling; as values approach zero, the gradient magnitude can explode, destabilizing the entire attention mechanism. The manuscript frankly admits this fragility in Section 5.5, noting that symmetric degree normalization compresses the adjacency weight range excessively, causing the log-barrier to become overly aggressive and destroying the router's discriminative capacity.

This catastrophic interaction between standard graph normalization techniques and the log-barrier mask indicates a fundamental architectural weakness. An architecture that breaks when subjected to ubiquitous graph preprocessing steps requires structural redesign, not merely hyperparameter avoidance. The authors should consider introducing an adaptive, learned temperature scaling parameter within the log-barrier equation: $\log(\tau \tilde{A}_{ij} + \epsilon)$. This would insulate the barrier from the compression effects of degree normalization, ensuring numerical stability and broadening the model's applicability across arbitrary graph datasets.

4. The Mixture-of-Experts Router for Multi-View Adjacencies

The deployment of a Mixture-of-Experts (MoE) routing paradigm over structural graph views represents the core operational engine of THORN. The authors must rigorously defend the design choices within this routing network, particularly concerning capacity allocation and parameter sharing.

4.1 Router Capacity, Specialization, and Load Balancing

The routing network in THORN utilizes a top-k gated softmax operating over the concatenated geometric observer features. The deliberate choice of $k=2$ active views out of $M=3$ total views allows each attention head to specialize in a complementary pair of views while maintaining computational sparsity. This sparse activation is a hallmark of modern MoE architectures, ensuring that inference costs do not scale linearly with the addition of new experts.

To prevent the well-documented pathology of expert collapse—where the gating network lazily routes all tokens to a single, dominant expert—THORN employs two critical auxiliary regularizers. The load balancing loss ($\mathcal{L}_{lb}$) penalizes uneven global view usage across the batch, while the head-view orthogonality loss ($\mathcal{L}_{orth}$) pushes the normalized Gram matrix of routing vectors toward a scaled identity matrix. This actively forces different attention heads to specialize on different views. Empirical analysis confirms this success: Head 0 focuses on geometry, Head 1 is kNN-dominant, and Head 2 specializes exclusively on temporal proximity. This discrete, interpretable specialization validates the efficacy of the geometric observer inputs.

4.2 The Parameter Sharing Bottleneck in Shared Q/K/V Projections

A rigorous comparative analysis reveals a severe representational bottleneck in THORN's current MoE implementation. In standard Large Language Model (LLM) MoEs, experts are massive, independent feed-forward networks possessing separate, unshared parameter weights. In stark contrast, the "experts" in THORN are merely different topological adjacency matrices that all pass through a single, shared set of Query, Key, and Value (Q/K/V) feature projections.

Forcing a single projection space to simultaneously process semantic feature relationships (via kNN edges) and temporal dynamics (via temporal edges) creates an inevitable, mathematical representation conflict. The linear transformations that optimally project features for spatial similarity will fundamentally contradict the transformations required to assess temporal proximity. The manuscript acknowledges this limitation, suggesting view-specific query-key projections as "future work". However, from an editorial perspective, this feature is critical to realizing the full potential of the MoE paradigm. Implementing separate $W_Q^{(m)}$, $W_K^{(m)}$, $W_V^{(m)}$ weight matrices for each graph view m would exponentially increase the expressive capacity of the architecture and properly align it with the established definition of a Mixture-of-Experts model.

5. Formalizing the Isolation-Versus-Interaction Tradeoff

The most profound theoretical contribution of the THORN manuscript is the formal identification and mathematical characterization of the isolation-versus-interaction tradeoff in multi-view attention. This complex dynamic is rarely addressed in contemporary multi-view learning literature, which generally defaults to simplistic additive view mixing, fixed heuristics, or static concatenation. By formalizing this tradeoff, THORN establishes a new theoretical framework for evaluating multi-source information propagation.

5.1 Pre-Softmax Mixing: Cross-View Synergy and Attention Dilution

In the Pre-softmax mixing formulation, colloquially termed the "Interaction Mode," the MoE router dynamically mixes the adjacency views before the attention softmax is applied over the entire union neighborhood. This mathematical sequence allows for explicit cross-view synergy. If an edge exists in multiple complementary views, the routing weights accumulate additively within the log-barrier, creating a natural, structurally reinforced signal.

This mode is highly effective for discovering cross-temporal or cross-spatial correspondences that no single view could resolve independently. However, this synergy comes at the severe cost of attention dilution. Because the softmax function must normalize over the vast union of all neighborhoods, dense and noisy views compete directly for probability mass against sparse, high-fidelity views. As the authors astutely analyze, residual edges from noisy views maintain non-zero weights through the log-barrier term, actively injecting mathematical noise into the final attention distribution and artificially capping performance.

5.2 Post-Softmax Mixing: Signal Isolation in High-SNR Regimes

Conversely, the Post-softmax mixing formulation operates as an "Isolation Mode." Here, each

graph view computes its own independent attention distribution using a view-local softmax, and the router subsequently combines the finalized attention outputs. This architectural sequence normalizes the probability mass exclusively within each view's specific neighborhood.
This isolation ensures that a dense, heavily corrupted view (e.g., a kNN graph driven by noise dimensions) cannot mathematically dilute the attention mass of a clean, highly predictive view (e.g., temporal proximity). By multiplicatively suppressing the noisy views post-softmax via low routing weights, the dominant signal is preserved at near-perfect fidelity. The unavoidable cost of this absolute isolation is the total loss of cross-view structural reinforcement; the network treats each topology as an impenetrable silo, failing to reward edges that demonstrate multi-view consensus.

5.3 Empirical Evidence and Regime Shifts in Synthetic Drift

The synthetic drift-strength experiment provided in the manuscript perfectly illustrates this theoretical divide, yielding compelling empirical ground support for the authors' claims. At a moderate drift level of 0.50, where the temporal view provides a near-perfect classification signal, post-softmax isolation dominates, achieving a staggering 0.984 PR-AUC. It effectively quarantines the noise introduced by the feature-based views.
However, at a drift level of 0.75, the data undergoes a severe regime shift, creating a scenario where temporal proximity alone becomes highly unreliable. In this complementary regime, pre-softmax interaction dominates, achieving 0.939 PR-AUC. The additive log-barrier accumulation identifies cross-temporal patterns that post-softmax's view-isolated attention is structurally blind to. The formal characterization provided by the authors—relying on the view overlap ratio \rho_i and the individual view Signal-to-Noise Ratio (SNR)—establishes a rigorous, predictive framework for routing architecture. This formalization proves that the optimal attention mode is an intrinsic property of the data's view correlation structure, not a mere implementation artifact. This is a high-impact, paradigm-building insight that must be adopted as a standard evaluation metric in future multi-view learning research.

Data Regime Configuration	Optimal THORN Mixing Mode	Mathematical Mechanism of Dominance
Drift \le 0.50 (Single Dominant View)	**Post-softmax** (Isolation)	Multiplicatively suppresses noisy views; prevents softmax denominator dilution.
Drift = 0.75 (Complementary Views)	**Pre-softmax** (Interaction)	Additive log-barrier accumulates weight on shared edges; exploits synergistic overlap.
Drift \ge 1.50 (Extreme Feature Noise)	**Time-only Baseline**	Multi-view routing overhead injects parameters without corresponding signal gain.

6. Cross-View Alignment via Fused Gromov-Wasserstein Transport

To enforce structural and semantic consistency across the routed views, THORN incorporates a Fused Gromov-Wasserstein (FGW) alignment regularizer. While the theoretical application of optimal transport here is elegant, the computational realities of this method present the most

significant operational vulnerability to the THORN framework.

6.1 Optimal Transport for Topological and Semantic Consistency

The Fused Gromov-Wasserstein distance provides a mathematically principled optimal transport framework for comparing heterogeneous metric-measure spaces. Unlike standard Wasserstein distances that require aligned feature spaces, FGW jointly considers feature distances (e.g., Euclidean node embeddings) and topological distances (e.g., shortest-path or diffusion distances) to compute a comprehensive structural alignment.

In the THORN architecture, the resulting FGW transport plans between differing views are cached as edge consensus features, which subsequently inform the per-head attention bias. Furthermore, a per-node discordance score is utilized within the alignment regularizer ($\mathcal{L}_{align}$) to actively penalize the MoE router for placing probability mass on views that exhibit high topological discordance. This mechanism ensures that the dynamic routing does not arbitrarily oscillate between wildly divergent topologies without a strong, feature-driven mathematical justification, thereby preventing negative transfer between views.

6.2 The Scalability Bottleneck and Offline Caching Limitations

Despite its theoretical brilliance, the computational complexity of FGW poses a catastrophic bottleneck. The calculation of exact Gromov-Wasserstein distances requires solving a complex, non-convex quadratic assignment problem. Even when mitigated by entropic regularization via the Sinkhorn algorithm and Frank-Wolfe optimization, the computational complexity scales prohibitively at $O(N^3)$ or $O(N^4)$.

To circumvent this intractable training time, the authors compute the FGW transport offline on ego-neighborhoods and statically cache the results. While this engineering compromise allows the model to train, it fundamentally breaks the architecture's applicability to dynamic or inductive settings. If the graph topology updates dynamically (e.g., streaming financial transactions), or if the model operates in an inductive environment where unseen nodes continuously arrive, the offline FGW cache becomes instantly deprecated. Recalculating $O(N^3)$ transport plans on the fly during inference is computationally impossible, rendering THORN entirely static in its current iteration.

6.3 Pathways to Online Approximations

To elevate THORN from a theoretically interesting static framework to a universally applicable, dynamic architecture, the authors must urgently address this transport alignment bottleneck. The authors are strongly advised to search academic databases for recent advancements in linear-time optimal transport.

Recent literature has demonstrated highly efficient approximations of the GW distance, such as the Graph-on-Logits Distillation (GLD) method, which reduces the complexity from $O(N^4)$ to $O(N \log N)$ while providing provable error bounds. Alternatively, linear-time Sinkhorn divergences utilizing low-rank couplings could be integrated directly into the forward pass. By adopting an $O(N \log N)$ approximation, THORN could compute cross-view consistency online, allowing the alignment regularizer to adapt dynamically to evolving graph structures. This modification is non-negotiable for deployment in enterprise-scale AI/ML environments.

7. Comparative Analysis with Established Architectures

To validate THORN's position and claims of paradigm advancement within the broader GNN ecosystem, it is absolutely necessary to rigorously compare it against established baseline models. The theoretical proofs of architectural subsumption provided in Theorem 2 must also be critically evaluated.

7.1 Subsumption of Graph Attention Networks (GAT/GATv2)

GAT and its dynamic successor, GATv2, introduced the foundational concept of masked self-attention to irregular graph topologies. Both baseline models compute attention coefficients based entirely on the semantic content of the source and target nodes, utilizing a single, rigid adjacency matrix to enforce structural sparsity.

THORN fundamentally departs from, and subsequently dominates, the GAT paradigm. As demonstrated in the manuscript's empirical results, GAT tops out at a 0.922 PR-AUC, systematically failing to match THORN across all temporal drift regimes. GAT suffers from profound "geometric blindness"—it cannot mathematically distinguish between an edge representing a tight intra-cluster bond and an edge acting as a fragile inter-cluster bridge, because both are treated equally by the non-differentiable binary mask. THORN's edge biases, derived from geometric observers like ORC, natively encode this critical geometric distinction.

Furthermore, Theorem 2 provides a rigorous mathematical proof: if THORN is restricted to a single view, with routing probabilities fixed to 1 and edge biases set to 0, the log-barrier formulation precisely recovers the standard masked neighborhood attention of GAT. Thus, THORN is proven to be a strict mathematical generalization of the GAT framework.

7.2 Contrasting with Graphormer and Global Attention Mechanisms

Graphormer and general Graph Transformers like GPS address the severe limitations of localized message-passing GNNs by utilizing fully connected global attention, augmented with fixed structural encodings such as shortest-path distances and degree centrality. This global perspective allows the model to capture long-range dependencies that are inevitably truncated in standard sparse GNNs.

THORN aligns closely with the foundational Graphormer philosophy of injecting structural priors as additive attention biases. However, a critical divergence exists: Graphormer relies on static, predetermined encodings applied uniformly to a fully connected graph space. THORN, conversely, utilizes dynamic, learned, structure-aware routing over multiple sparse views.

Theorem 2 successfully sketches the reduction framework: if THORN operates on a fully connected graph view (rendering the log-barrier term a vanishing constant) and the edge observer features encode shortest paths, it perfectly mimics the Graphormer architecture.

THORN’s massive comparative advantage over Graphormer is its computational scalability; by routing attention dynamically over sparse views via MoE, THORN bypasses the crippling O(N^2) quadratic bottleneck of global attention while maintaining the capacity to access distant nodes via views like multi-scale diffusion.

7.3 Alignment with Geometric Mixture-of-Experts

The concept of applying Mixture-of-Experts (MoE) to geometric deep learning is nascent but rapidly evolving, particularly in the domains of molecular property prediction and physics-informed neural networks. Frameworks such as SPiCE and MI-MoE utilize Geometric Mixture-of-Experts (GMoE) to selectively route atomic coordinates and complex scalar/vector features to highly specialized 3D and 2D GNN backbones. Similarly, the Curvature-Adaptive Transformer (CAT) dynamically routes nodes to different Riemannian curvature spaces for optimized processing.

Compared to these contemporary GMoE architectures, THORN takes a uniquely structural, rather than feature-based, approach. While SPiCE routes different feature types to different expert networks , and CAT routes nodes to different continuous geometric spaces , THORN routes attention heads to entirely different discrete adjacency structures. THORN's experts are not distinct neural networks laden with separate parameter weights; they are distinct topological graphs sharing a common projection space. This unique distinction renders THORN significantly more parameter-efficient than standard MoEs, as it only scales the adjacency processing overhead rather than duplicating massive feed-forward blocks.

Architectural Framework	Routing Mechanism	View / Topology Assumption	Core Structural Bias Integration	Computational Complexity
GAT / GATv2	None (Static Mask)	Single, fixed graph	None (Content-only attention)	O(E \cdot d)
Graphormer	None (Global Attn)	Single, fully-connected	Shortest-path, Degree biases	O(N^2 \cdot d)
SPiCE (GMoE)	Feature-based MoE gating	Dynamic cutoff graphs	3D invariant/equivariant features	O(E \cdot d_{expert} \cdot k)
THORN	Geometric Observer MoE gating	Multiple routed views	ORC, LID, LOF, FGW consensus	O(E_{un[span_2](start_span)[span_2](end_span)ion} \cdot d) + O_{FGW_{offline}}

8. Specific Challenges and Architectural Vulnerabilities

While THORN demonstrates exceptional theoretical maturity and mathematical elegance, a rigorous editorial critique must ruthlessly highlight the empirical and architectural vulnerabilities that currently limit the framework's broader applicability in the field.

8.1 Sensitivity to Graph Normalizations

Section 5.5 of the manuscript frankly admits that several mathematically motivated graph operations—specifically symmetric degree normalization and the addition of self-loops—actively degrade THORN's empirical performance. The addition of self-loops dropped the PR-AUC by 4.1%, while row normalization caused a catastrophic 9.1% collapse in predictive power. The authors correctly diagnose the root cause: the soft Gaussian kernel weights utilized in THORN's

multi-view construction are excessively compressed by symmetric normalization, which then interacts catastrophically with the log-barrier mask, destroying the router's ability to discriminate between views.

While this diagnostic reasoning is sound, the resulting empirical fragility is deeply concerning. An architecture that fundamentally breaks when subjected to the most common, ubiquitous graph normalization techniques requires structural redesign. The model must be capable of processing diverse datasets without requiring bespoke, fragile preprocessing pipelines.

8.2 Data Requirements and the Insufficiency of Synthetic Benchmarks

The empirical evidence provided in the manuscript is restricted entirely to a highly controlled, synthetic dataset consisting of merely 512 nodes with 16-dimensional features. While synthetic benchmarks are excellent for isolating specific mechanistic behaviors—such as the highly elegant demonstration of the temporal drift-strength tradeoff—they are wholly insufficient for proving the efficacy of a paradigm-shifting architecture.

The complexities, noise profiles, and massive scales of real-world graphs (e.g., severe heterophily, power-law degree distributions, highly missing features) often cause theoretically sound architectures to collapse entirely in practice. To meet the exacting standards of a premier journal, THORN must be rigorously evaluated on large-scale, standard industry benchmarks such as the Open Graph Benchmark (OGB). Specifically, dynamic datasets like ogbn-arxiv or massive molecular datasets like PCQM4Mv2 are required to validate the scalability and robustness of the geometric observer routing.

9. Editor's Directives for High-Impact Publication

To elevate the THORN framework from a highly promising theoretical blueprint to a definitive, high-impact publication capable of passing peer review in venues of JMLR or Nature Machine Intelligence caliber, the following structural and empirical refinements must be executed:

1. **Scale the Empirical Validation:** The immediate, non-negotiable priority is applying THORN to massive, real-world graph datasets. The profound isolation-versus-interaction tradeoff must be demonstrated on authentic multiplex networks to prove that the pre/post-softmax dynamics hold true outside of carefully constructed synthetic drift environments.
2. **Resolve the Optimal Transport Bottleneck:** Replace the exact offline Fused Gromov-Wasserstein calculation with an online, linear-time approximation. Exploring Graph-on-Logits Distillation (GLD) or low-rank Sinkhorn factorizations will allow THORN to function continuously on dynamic graphs, drastically expanding its industrial utility and theoretical completeness.
3. **Implement View-Specific Projections:** The authors must instantiate separate W_Q, W_K, W_V weight matrices for each graph view to resolve the representation conflict inherent in shared projections. The MoE routing probabilities should dynamically modulate not just the adjacency matrix, but the continuous representation spaces themselves, unlocking the true capacity of the expert network.
4. **Stabilize the Log-Barrier Mechanism:** Introduce an adaptive, parameter-learned temperature scaling mechanism within the log-barrier equation. This modification is critical to insulate the continuous relaxation from the catastrophic compression effects of standard graph normalization, rendering the architecture robust and plug-and-play for

arbitrary datasets.

THORN establishes a rigorous, mathematically sophisticated framework that directly addresses the topological blindness of legacy graph attention models. With targeted refinement addressing its computational bottlenecks and empirical scope, the architecture possesses the theoretical profoundness to fundamentally alter the trajectory of representation learning on multiplex graphs.

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